Structured Attribute-Guided Embeddings for Efficient Fashion Image Retrieval

Impact Statement

Fashion image retrieval systems are increasingly used in e-commerce, recommendation platforms, and visual search applications. However, many existing approaches require large-scale training and high computational cost, which can limit their practical use in resource-constrained settings.

This work presents a lightweight retrieval framework that builds on pretrained vision–language representations while training only a small projection network. The proposed structured embedding design helps organize clothing features more effectively within the representation space and improves retrieval behavior without requiring extensive hardware resources.

The experimental results show that meaningful retrieval performance can be achieved using modest compute and a relatively small dataset. Beyond fashion retrieval, the proposed idea of structured embeddings may also be useful for other multimodal retrieval problems where fine-grained visual similarity is important, such as product search, image recommendation, and visual catalog organization.